

Pneumonia Detection from Chest X-Ray Images Using Deep Learning

Deep Learning Project Report

Binary Image Classification: NORMAL vs. PNEUMONIA

Author: Alireza Shahidiani

Matricola: 0001154402

Dataset: Kaggle Chest X-Ray Images (Pneumonia)

Framework: PyTorch 2.10

Hardware: NVIDIA L4 GPU (Google Colab)

Language: Python 3.12

Abstract

This project investigates the application of deep learning techniques for automated pneumonia detection from chest X-ray images. Multiple architectures were developed and systematically compared, including a baseline Convolutional Neural Network (CNN), weighted CNN variants, and a fine-tuned ResNet50 transfer learning model. Class imbalance, model regularisation, and evaluation reliability were addressed through weighted loss functions, label smoothing, K-Fold Cross Validation, and Grad-CAM interpretability analysis. The final production model achieved a test accuracy of **85%** and a weighted F1-score of **0.84**, with NORMAL recall improving from 0.32 in the baseline to 0.64 in the final model. The study also identifies and corrects a validation augmentation leakage issue present in early experimentation, and critically evaluates the substantial gap between K-Fold validation performance and held-out test performance in the context of known dataset distribution differences.

Contents

1	Introduction	3
1.1	Objectives	3
2	Dataset Description and Preprocessing	3
2.1	Dataset Overview	3
2.2	Data Preprocessing and Augmentation	4
3	Model Architectures	5
3.1	Custom CNN Architecture	5
3.2	Fine-Tuned ResNet50	5
4	Training Methodology	6
4.1	Addressing Class Imbalance with Weighted Loss	6
4.2	Training Configuration	6
4.3	Label Smoothing in the Improved Model	6
5	Experimental Results	7
5.1	Baseline CNN	7
5.2	Weighted CNN	7
5.3	Improved Weighted CNN	7
5.4	Fine-Tuned ResNet50	8
6	K-Fold Cross Validation	8
6.1	Methodological Note: Augmentation Leakage	10
7	Final Production Model	10
8	Model Comparison	11
8.1	Analysis	12
9	Grad-CAM Interpretability	13
10	Conclusion	13

1. Introduction

Pneumonia is a serious pulmonary infection that affects millions of people worldwide and remains one of the leading causes of morbidity and mortality, particularly in young children and elderly populations. Early and accurate diagnosis is essential for effective treatment and improved patient outcomes. Chest X-ray imaging is the most widely used diagnostic tool for pneumonia detection; however, manual interpretation of radiological images is time-consuming, subjective, and highly dependent on the experience of the radiologist.

Deep learning has emerged as a powerful tool for automated medical image analysis. Convolutional Neural Networks (CNNs) have demonstrated remarkable performance in image classification tasks and have shown strong potential in clinical settings. Transfer learning, in particular, allows pre-trained models to be adapted to specialised domains such as medical imaging, even when labelled training data is limited.

This project develops and evaluates multiple deep learning models for binary classification of chest X-ray images into two categories: **NORMAL** and **PNEUMONIA**. The study systematically examines the impact of class imbalance, regularisation techniques, transfer learning, and cross-validation on model performance.

1.1 Objectives

The main objectives of this project are:

1. Develop a baseline CNN to establish initial performance on the classification task.
2. Address class imbalance using weighted cross-entropy loss and evaluate its impact.
3. Investigate regularisation improvements including label smoothing and learning rate tuning.
4. Compare custom CNN architectures against a fine-tuned ResNet50 transfer learning model.
5. Validate model robustness using 5-Fold Cross Validation.
6. Apply Grad-CAM to improve model interpretability and clinical transparency.
7. Train a final production model on all available non-test data for maximum performance.

2. Dataset Description and Preprocessing

2.1 Dataset Overview

The dataset used in this project is the publicly available *Chest X-Ray Images (Pneumonia)* dataset obtained from Kaggle [1]. The dataset contains chest X-ray images of

paediatric patients from Guangzhou Women and Children’s Medical Center. Images are labelled as either **NORMAL** or **PNEUMONIA**, with the latter encompassing both bacterial and viral pneumonia cases.

The dataset exhibits a significant class imbalance, with pneumonia cases outnumbering normal cases by approximately 3:1 in the training set. This imbalance is a fundamental challenge addressed throughout the project.

Table 1: Dataset split and class distribution.

Split	NORMAL	PNEUMONIA	Total
Training (full)	1,341	3,875	5,216
Experimental Train (80%)	1,058	3,114	4,172
Experimental Val (20%)	283	761	1,044
Test (held-out)	234	390	624

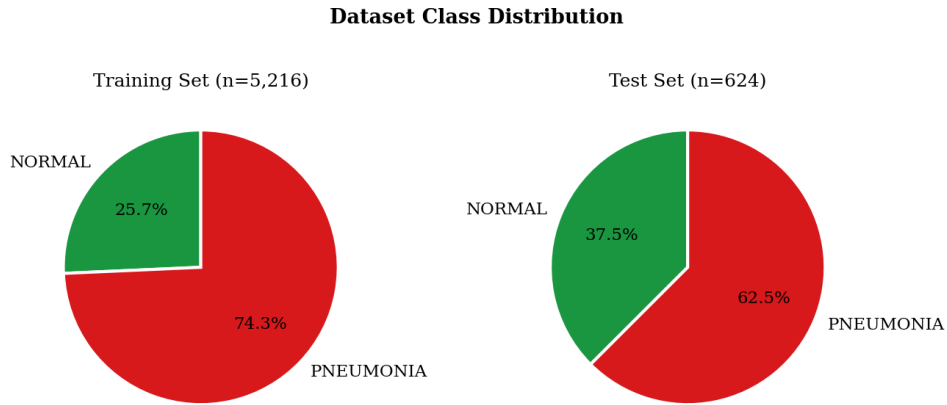


Figure 1: Class distribution in the training and test sets. The training data shows a notable imbalance with approximately 74% pneumonia cases.

2.2 Data Preprocessing and Augmentation

All images were preprocessed using a consistent pipeline. Since the original X-ray images are grayscale, they were converted to three-channel images to satisfy the input requirements of both the custom CNN and the pre-trained ResNet50.

Images were resized to 224×224 pixels and normalised using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$), which is appropriate for both the custom CNN and the ResNet50 fine-tuning task.

Training images were augmented with **RandomHorizontalFlip** and **RandomRotation** ($\pm 10^\circ$) to improve model robustness to spatial variations in X-ray acquisition. Validation and test transforms consisted of deterministic resize and normalisation only, with no augmentation applied.

To prevent the **validation data leak** that would occur if both splits shared the same augmented dataset object, two separate **ImageFolder** instances were created from the

same root directory — one with `train_transform` and one with `val_transform` — and then subsetting using shared random indices.

3. Model Architectures

3.1 Custom CNN Architecture

A custom four-block Convolutional Neural Network was developed as the primary architecture. The network was designed to progressively extract spatial features from the input image through increasingly complex convolutional representations.

Table 2: PneumoniaCNN architecture summary.

Layer	Type	Output Size	Parameters
Input	—	$3 \times 224 \times 224$	—
Block 1	Conv2d(3→32) + BN + ReLU + MaxPool	$32 \times 112 \times 112$	896
Block 2	Conv2d(32→64) + BN + ReLU + MaxPool	$64 \times 56 \times 56$	18,496
Block 3	Conv2d(64→128) + BN + ReLU + MaxPool	$128 \times 28 \times 28$	73,856
Block 4	Conv2d(128→256) + BN + ReLU + MaxPool	$256 \times 14 \times 14$	295,168
Flatten	—	50,176	—
FC1	Linear(50176→512) + ReLU + Dropout(0.5)	512	25,690,624
FC2	Linear(512→2)	2	1,026
Total trainable parameters			26,080,066

Each convolutional block follows the pattern: `Conv2d` → `BatchNorm2d` → `ReLU` → `MaxPool2d(2)`. Batch normalisation was included in each block to stabilise training and reduce sensitivity to the weight initialisation. A Dropout layer with probability 0.5 was placed before the final classification head to reduce overfitting.

3.2 Fine-Tuned ResNet50

In addition to the custom CNN, a pre-trained ResNet50 model was fine-tuned for the binary classification task. ResNet50’s residual connections (skip connections) help address the vanishing gradient problem in deep networks, allowing the model to learn more discriminative features.

The transfer learning strategy was as follows:

- All pre-trained layers were frozen to preserve ImageNet feature representations.
- Only `layer4` (the final residual block) was unfrozen for domain adaptation.
- The original 1,000-class fully connected head was replaced with a custom binary classifier: `Linear(2048→256)` → `ReLU` → `Dropout(0.5)` → `Linear(256→2)`.

A very small learning rate (5×10^{-5}) was used to fine-tune the unfrozen layers without disrupting the pre-trained ImageNet representations. A `ReduceLROnPlateau` scheduler with factor 0.5 and patience 1 was also applied to stabilise convergence.

4. Training Methodology

4.1 Addressing Class Imbalance with Weighted Loss

The dataset’s 3:1 imbalance between pneumonia and normal cases causes unweighted models to predict PNEUMONIA for nearly all inputs, producing high overall accuracy but a rate of false positive pneumonia predictions that is undesirable for practical deployment. Class weights were computed using the standard formula:

$$w_c = \frac{N_{\text{total}}}{2 \times N_c} \quad (1)$$

where N_{total} is the total number of training samples and N_c is the number of samples in class c . For the experimental split, this yielded $w_{\text{NORMAL}} = 1.97$ and $w_{\text{PNEUMONIA}} = 0.67$, effectively up-weighting the minority normal class in the loss computation.

4.2 Training Configuration

The general `train_model` function was used for all experimental models (baseline, weighted CNN, improved weighted CNN, and ResNet50). Key configuration details:

Table 3: Training hyperparameters across experimental models.

Setting	Baseline	Weighted	Improved	ResNet50
Optimiser	Adam	Adam	Adam	Adam
Learning Rate	0.001	0.001	0.0007	0.00005
Batch Size	32	32	32	32
Max Epochs	20	20	20	20
Early Stop Patience	3	3	3	4
Loss Function	CE	Weighted CE	Weighted CE + LS	Weighted CE
LR Scheduler	None	None	None	ReduceLROnPlateau

Early stopping monitored validation loss and halted training when no improvement was observed for p consecutive epochs (patience). **Model checkpointing** saved the best-performing model state to disk at each improvement, ensuring the loaded model corresponds to the lowest validation loss, not the final epoch.

4.3 Label Smoothing in the Improved Model

For the improved weighted CNN, label smoothing with $\epsilon = 0.1$ was added to the weighted cross-entropy loss. Label smoothing softens the one-hot target distribution by mixing it with a uniform distribution:

$$\tilde{y}_c = (1 - \epsilon) y_c + \frac{\epsilon}{K} \quad (2)$$

where $K = 2$ is the number of classes. This reduces model overconfidence and improves calibration and generalisation, particularly in scenarios with limited training data.

5. Experimental Results

5.1 Baseline CNN

The baseline CNN was trained with standard (unweighted) cross-entropy loss to establish the extent of class-imbalance bias. Early stopping triggered at epoch 5. On the held-out test set, the model achieved 74% accuracy but showed extreme bias toward the PNEUMONIA class, with NORMAL recall of only 0.32. This means 159 out of 234 healthy patients were incorrectly classified as having pneumonia — a result undesirable for practical deployment that clearly motivates the need for class-balancing strategies.

Table 4: Baseline CNN classification report on the test set (n=624).

Class	Precision	Recall	F1-Score	Support
NORMAL	0.97	0.32	0.48	234
PNEUMONIA	0.71	0.99	0.83	390
Weighted avg	0.81	0.74	0.70	624

5.2 Weighted CNN

Introducing weighted cross-entropy loss substantially improved classification balance. Early stopping triggered at epoch 13. NORMAL recall improved from 0.32 to 0.44, indicating that the class weights successfully penalised misclassification of the minority class. PNEUMONIA recall remained very high at 0.98.

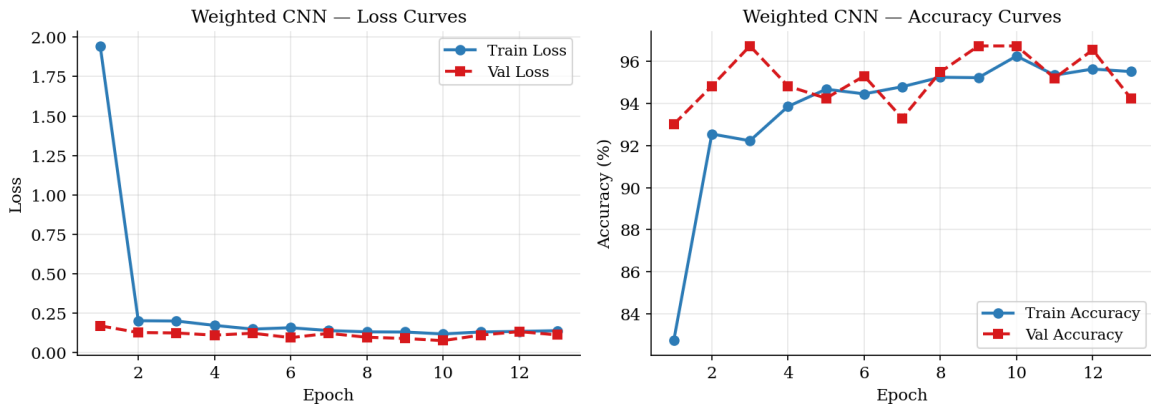


Figure 2: Weighted CNN training and validation loss (left) and accuracy (right) over 13 epochs. Validation loss converges steadily without divergence, indicating stable learning behaviour.

5.3 Improved Weighted CNN

The improved model added label smoothing ($\epsilon = 0.1$) and a reduced learning rate (0.0007). Early stopping triggered at epoch 12. These additions produced a significant performance gain: test accuracy rose to 83% and weighted F1 to 0.81. NORMAL recall increased further to 0.57, representing a substantial reduction in false pneumonia predictions for healthy patients.

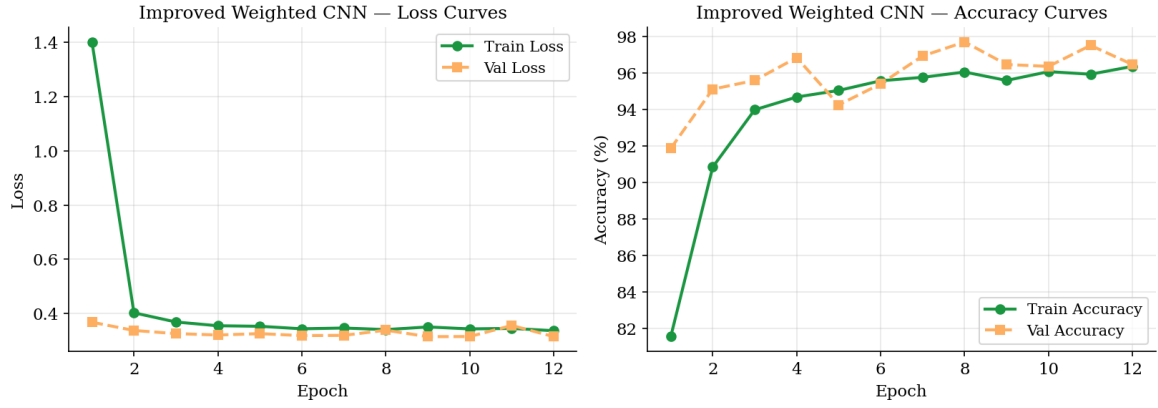


Figure 3: Improved Weighted CNN training dynamics. The elevated but stable loss values relative to the basic weighted CNN are expected due to label smoothing, which intentionally reduces loss magnitude. Both train and validation curves converge smoothly without divergence.

5.4 Fine-Tuned ResNet50

The fine-tuned ResNet50 achieved the strongest performance among all experimental models. Training converged rapidly due to the pre-trained ImageNet weights, reaching near-peak performance within 5 epochs. Early stopping triggered at epoch 12. The model achieved 85% test accuracy, weighted F1 of 0.84, and NORMAL recall of 0.61 — the highest among all experimental models at that stage.

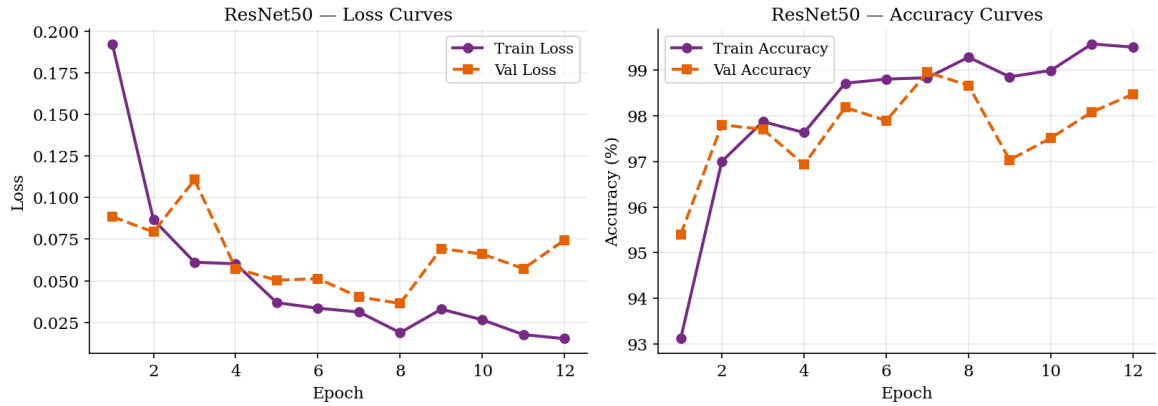


Figure 4: ResNet50 training dynamics. The model converges rapidly with very low loss values, demonstrating the strong feature representations provided by ImageNet pre-training. Minor validation loss fluctuations in later epochs are typical of confidence-based overfitting without meaningful accuracy degradation.

6. K-Fold Cross Validation

To assess the robustness and generalisability of the weighted CNN, 5-Fold Cross Validation was performed on the full training set (5,216 images). K-Fold evaluation reduces dependence on any single train-validation partition and provides a more statistically reliable estimate of model performance.

Implementation details:

- Two separate `ImageFolder` instances were created — one with `train_transform` and one with `val_transform` — to prevent augmentation from leaking into validation folds.
- Class weights were recomputed independently for each fold’s training subset.
- A fresh model was initialised at the start of each fold.
- Early stopping (patience 2), checkpointing, and 10 maximum epochs were used per fold.

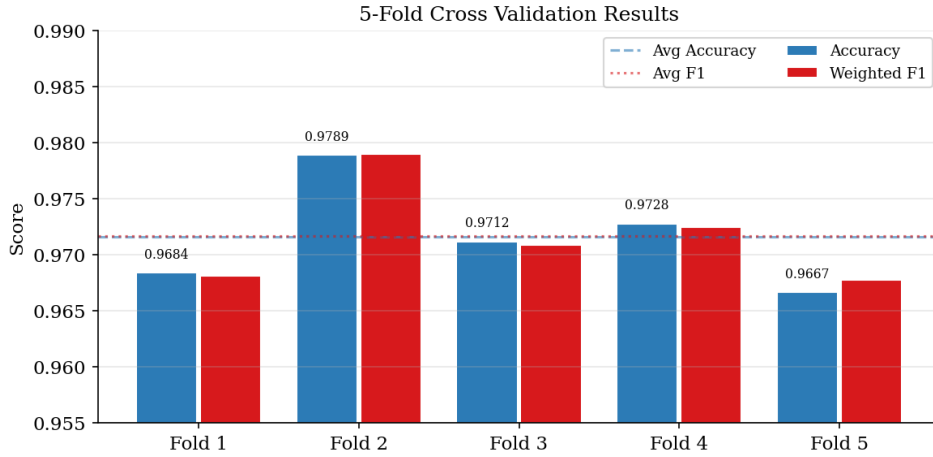


Figure 5: 5-Fold Cross Validation accuracy and weighted F1 scores per fold, with dashed lines indicating the overall averages. The low variance across folds confirms model stability.

Table 5: 5-Fold Cross Validation summary results.

Metric	Mean	Std Dev
Accuracy	0.9716	0.0046
Weighted F1	0.9717	0.0046

The model achieved an average accuracy of **97.16%** and average weighted F1 of **97.17%** across all five folds. The very low standard deviation (≈ 0.005) demonstrates that the model performs consistently regardless of the particular data split chosen. These results confirm the robustness of the weighted CNN training strategy and validate its suitability for the final production retraining phase.

The gap between the K-Fold validation accuracy ($\sim 97\%$) and the held-out test accuracy ($\sim 83\%$) is likely attributable, at least in part, to distribution differences between the original training and independent test folders. This is a well-documented characteristic of this particular Kaggle dataset, where training and test images were reportedly collected under different acquisition conditions. Additional contributing factors may include the relatively small test set size (624 images) and the possibility that the training distribution does not fully represent the variability present in the test set. The

held-out test result therefore remains the primary benchmark for assessing real-world generalisation performance.

6.1 Methodological Note: Augmentation Leakage

During early experimentation, a validation augmentation leakage issue was identified. The initial implementation used a single `ImageFolder` dataset with training transforms, which was then split into training and validation subsets. As a result, validation images were evaluated with random augmentations applied — introducing stochasticity into the evaluation loop and artificially inflating validation performance estimates.

This issue was corrected by constructing two separate `ImageFolder` instances from the same root directory: one with `train_transform` (including flipping and rotation) and one with `val_transform` (deterministic only). The same index partition was then applied to both, ensuring that training and validation subsets are drawn from the same split while receiving appropriate transforms. The same correction was applied to K-Fold validation folds. All results reported in this document reflect this corrected implementation.

7. Final Production Model

After completing architecture selection and validation experiments, a final production model was trained on the *entire* non-test dataset (5,216 images). This approach is standard practice in machine learning: once hyperparameters and architecture have been finalised, all available labelled data should be used to maximise feature learning before the single, definitive test evaluation.

Since no validation set was held out during this stage, the `train_final_model` function was used — a simplified training loop without early stopping. The number of epochs (12) was selected based on the convergence patterns observed in the earlier experiments. Weighted cross-entropy loss with Adam ($\text{lr} = 0.001$) was used.

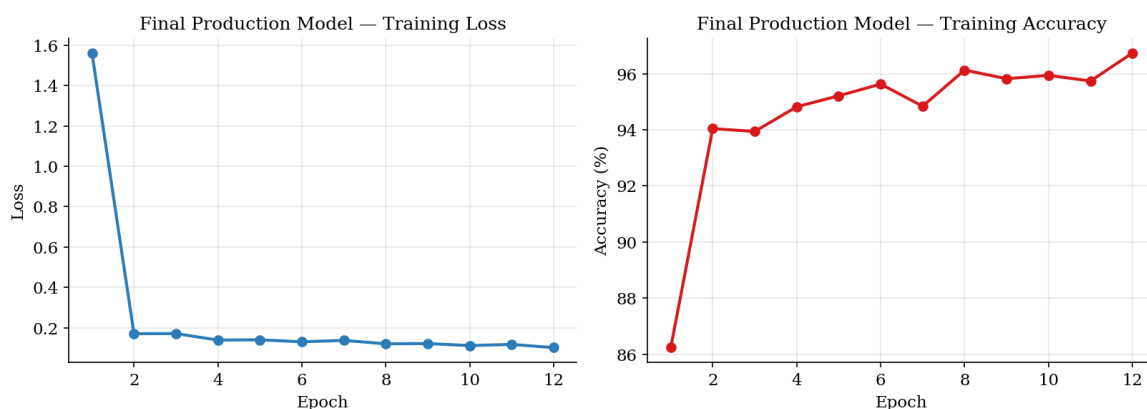


Figure 6: Final production model training loss (left) and accuracy (right) over 12 epochs. The loss decreases steadily and accuracy approaches 97%, indicating successful convergence on the full dataset.

The final model was evaluated once on the held-out test set (624 images). It achieved a test accuracy of **85%**, weighted F1 of **0.84**, and — critically — NORMAL recall of **0.64**: the highest healthy-patient recognition rate of any model in the project. PNEUMONIA recall remained strong at 0.97.

Table 6: Final Production Model classification report on the test set (n=624).

Class	Precision	Recall	F1-Score	Support
NORMAL	0.93	0.64	0.76	234
PNEUMONIA	0.82	0.97	0.89	390
Weighted avg	0.86	0.85	0.84	624

8. Model Comparison

Table 7: Comprehensive test-set performance comparison across all models.

Model	Accuracy	Weighted F1	NORMAL Recall	PNEUMONIA Recall
Baseline CNN	0.74	0.70	0.32	0.99
Weighted CNN	0.78	0.76	0.44	0.98
Improved Weighted CNN	0.83	0.81	0.57	0.98
Fine-Tuned ResNet50	0.85	0.84	0.61	0.99
Final Production CNN	0.85	0.84	0.64	0.97

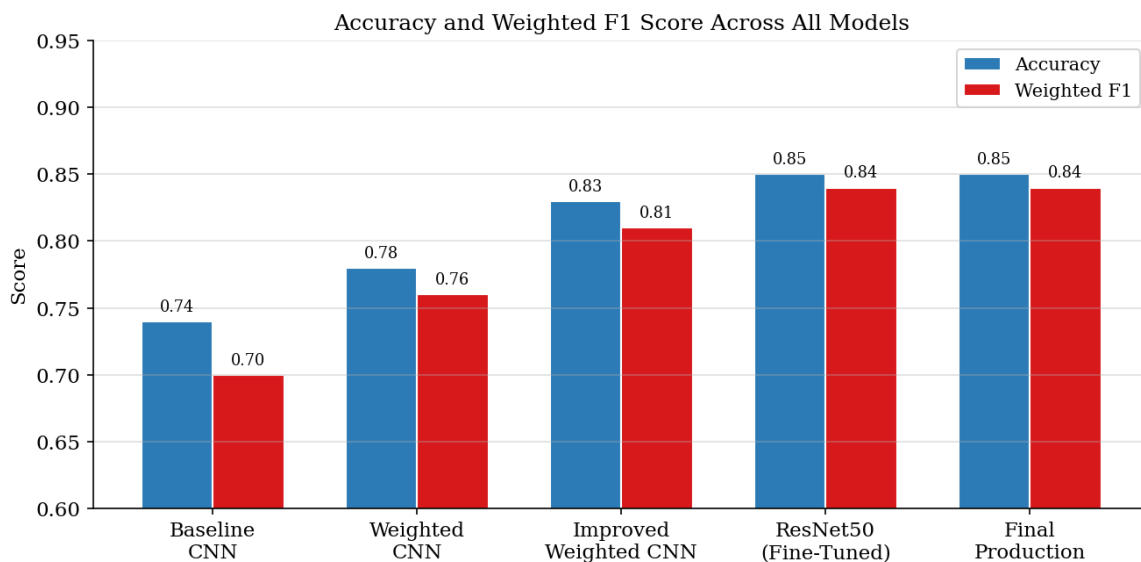


Figure 7: Accuracy and Weighted F1 score comparison across all five models. A clear and monotonic improvement is observed from the baseline CNN to the final production model.

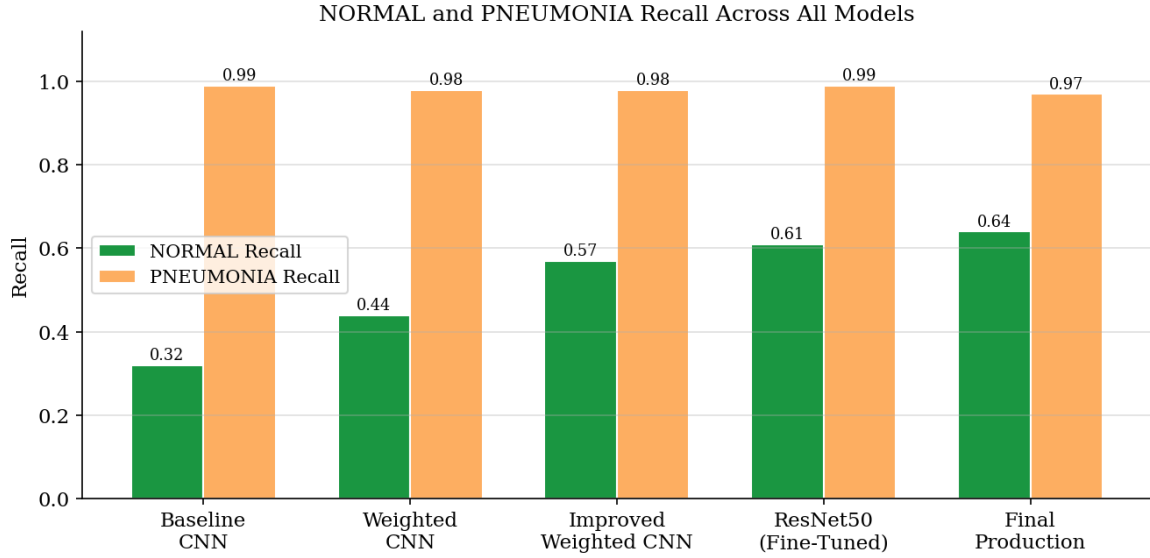


Figure 8: NORMAL and PNEUMONIA recall comparison. The NORMAL recall improvement from 0.32 (baseline) to 0.64 (final production) illustrates the progressive effect of class weighting, regularisation, and production retraining on minority-class recognition.

8.1 Analysis

The results in Table 7 and Figures 7–8 reveal several important findings:

Impact of class weighting: The introduction of weighted loss produced the single largest NORMAL recall improvement in the study, raising it from 0.32 to 0.44. This confirms that unweighted training on imbalanced medical datasets produces systematically biased classifiers with a rate of false positive pneumonia predictions that is undesirable for practical deployment.

Value of regularisation: Adding label smoothing and a lower learning rate (improved weighted CNN) provided a further substantial NORMAL recall gain (+0.13) over the standard weighted CNN, demonstrating that reducing model overconfidence is particularly beneficial in medical imaging contexts.

Effectiveness of transfer learning: ResNet50’s pre-trained ImageNet representations enabled superior feature extraction, achieving 85% accuracy and 0.61 NORMAL recall — matching the final production CNN’s accuracy and outperforming all custom CNNs except the final production model. The rapid convergence (stable by epoch 5) demonstrates the efficiency of transfer learning for small medical imaging datasets.

Production retraining benefit: Training the final model on all non-test data provided a meaningful NORMAL recall gain (+0.03) over the ResNet50 experimental result, at matched accuracy and F1, confirming that maximising training data is effective when architecture and hyperparameters are already validated.

Precision–Recall tradeoff: As NORMAL recall improved progressively across experiments (from 0.32 to 0.64), a corresponding gradual reduction in PNEUMONIA recall was observed (from 0.99 to 0.97). This reflects the inherent tradeoff between

reducing false positive pneumonia predictions for healthy patients and maintaining the highest possible sensitivity for true pneumonia cases. In clinical screening contexts, the relative cost of these two error types depends on downstream triage protocols and should be considered when selecting a deployment threshold.

9. Grad-CAM Interpretability

Deep learning models are often criticised as “black-box” systems, which is particularly problematic in clinical settings where model decisions must be transparent and auditable. Gradient-weighted Class Activation Mapping (Grad-CAM) [3] was applied to the final production model to visualise the image regions most influential in the model’s predictions.

Grad-CAM computes the gradient of the class score with respect to the final convolutional feature maps, then applies global average pooling to obtain per-channel importance weights. The weighted combination of the feature maps produces a localisation heatmap:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c \cdot A^k \right), \quad \alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (3)$$

where A^k is the activation of feature map k , y^c is the score for class c , and Z is the spatial dimension of the feature map.

The target layer was the final convolutional layer (`features[12]: Conv2d 128 → 256`), which captures the highest-level spatial features before the fully connected classifier.

The Grad-CAM analysis indicated that the model primarily attended to thoracic and lung parenchymal regions during classification. Heatmaps highlighted activation patterns in areas consistent with clinically relevant regions of the chest radiograph, suggesting that the model may be relying on clinically relevant thoracic regions during prediction, rather than peripheral image artefacts or acquisition metadata. While a single qualitative example is insufficient for definitive conclusions, this provides preliminary visual evidence that the model’s decisions are grounded in anatomically plausible image regions.

10. Conclusion

This project successfully developed and evaluated a series of deep learning models for automated pneumonia detection from chest X-ray images. The systematic experimental progression — from baseline CNN to weighted CNN variants, through transfer learning with ResNet50, validated by K-Fold Cross Validation, and finalised with production retraining — provides a complete and well-documented methodology for medical image classification.

Key findings:

1. **Class imbalance is a critical challenge.** Without class weighting, models achieve superficially high accuracy by predicting PNEUMONIA almost universally, producing undesirably low NORMAL recall (0.32) and a high rate of false positive pneumonia predictions.
2. **Weighted loss is highly effective.** Class weighting delivered the largest single recall improvement across all interventions evaluated in this study.
3. **Regularisation further improves class balance.** Label smoothing and reduced learning rate produced a further +0.13 NORMAL recall gain, reducing model overconfidence.
4. **Transfer learning accelerates convergence and improves performance.** ResNet50 achieved superior accuracy with significantly faster convergence, confirming the value of pre-trained representations for small medical imaging datasets.
5. **K-Fold validation confirms model robustness.** The very low standard deviation across folds ($\sigma \approx 0.005$) demonstrates that the weighted CNN generalises consistently and does not depend on any particular data split.
6. **Production retraining on all available data maximises final performance.** The final model achieved the strongest NORMAL recall (0.64) of any model evaluated.

The final production weighted CNN achieved the strongest NORMAL recall and the best overall class balance among all evaluated models, while the fine-tuned ResNet50 remained the strongest transfer-learning benchmark in terms of overall experimental performance, achieving matched accuracy and F1 with higher PNEUMONIA recall. Neither model dominates on all metrics, and the preferred choice depends on the relative cost assigned to false positive and false negative errors in the intended deployment context.

Limitations

This project also has several limitations that should be acknowledged. The dataset is relatively small and originates from a single public benchmark source, which may limit generalisability to real clinical environments with more heterogeneous patient populations and imaging equipment. The train and test folders exhibit noticeable distribution differences, contributing to the substantial gap observed between K-Fold validation and held-out test performance; this gap may reflect dataset-specific artefacts rather than fundamental model limitations. Additionally, the study focuses on binary classification and does not distinguish between bacterial and viral pneumonia subtypes, which carry different clinical implications. Finally, all reported results are based on a single test-set evaluation; without confidence intervals, the statistical significance of

differences between models with similar scores (e.g. ResNet50 vs. Final Production CNN) cannot be formally assessed.

Future Work

Future work could explore classification threshold optimisation as a post-hoc strategy to further improve NORMAL recall without retraining — adjusting the decision boundary away from the default argmax could shift the precision–recall tradeoff in favour of healthy-patient recognition at minimal cost to pneumonia sensitivity. Replacing the fully connected classifier with a Global Average Pooling (GAP) layer would substantially reduce the model’s parameter count from approximately 25.6 million to under 1,000 for that layer, potentially improving generalisation and reducing memory requirements. Ensemble approaches combining CNN and ResNet50 predictions may further improve robustness. Extending the study to multi-class classification distinguishing bacterial from viral pneumonia, and evaluating models on independent clinical cohorts, would strengthen the clinical relevance of the findings.

References

- [1] Mooney, P. (2018). *Chest X-Ray Images (Pneumonia)*. Kaggle. <https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>
- [2] He, K., Zhang, X., Ren, S., & Sun, J. (2016). *Deep Residual Learning for Image Recognition*. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 770–778.
- [3] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). *Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization*. IEEE International Conference on Computer Vision (ICCV), pp. 618–626.
- [4] Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., & Wojna, Z. (2016). *Rethinking the Inception Architecture for Computer Vision*. IEEE CVPR, pp. 2818–2826.
- [5] Paszke, A., et al. (2019). *PyTorch: An Imperative Style, High-Performance Deep Learning Library*. Advances in Neural Information Processing Systems (NeurIPS), 32.
- [6] Kohavi, R. (1995). *A Study of Cross-Validation and Bootstrap for Accuracy Estimation and Model Selection*. International Joint Conference on Artificial Intelligence (IJCAI), 14(2), pp. 1137–1145.